Note on Data Encryption

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– 1 – Introduction.

Modern computer science releis a lot on securety, either via phisical solutions or software ones. In this notes we focus on the latter; we mainly cover the most commonly used algorithm in cryptography, as well as some codes used for comunications.

In the remainder of this section we provide the reader with some basic definitions of abstract algebra. Later we Introduce the notion of elliptic curves and present some algorithm that make use of them.

– 1.1 – Notions of abstract algebra.

In this section, we first introduce the notion of group, ring and field, we then define what a field extension is and how to it is used. In describe what above, we assume the reader has a basic understanding of modulo arithmetics.

Definition (Group): Let G be a set, and let $*$ denote a binary operation on G's elements. We call *group* the pair $(G, *)$ if the following hold:

1. There exists a neutral element in G; that is,

$$\exists e \in G \text{ s.t. } \forall a \in G, a * e = e * a = a$$

2. Each element has an inverse, i.e,

$$\forall a \in G, \exists a^{-1} \text{ s.t. } a * a^{-1} = a^{-1} * a = e$$

3. The operation $*$ is associative, meaning that

$$\forall a, b, c \in G, (a * b) * c = a * (b * c)$$

Additionally, if $*$ is such that $a * b = b * a$, i.e., $*$ is commutative, we say that the group is *abelian*.

The number of elements of a group G is said to be its order $|G|$.

Definition (Cyclic group): Let $(G, *)$ be a group, and let $a \in G$. We denote by a^n the n-th power of a, and denote the set of all such powers by $\langle a \rangle$. We say that $(G, *)$ is cyclic if $\langle a \rangle = G$.

Definition (Sub-group): Given $(G, *)$ a group, we say that $H \subseteq G$ forms a sub-group under $*$, if $(H, *)$ is itself a group.

The following is one of the core theorems in group theory, and its extentions to rings, comes in handy when dealing with fields.

Theorem (Lagrange's theorem): *Let $(G, *)$ be a group and let $(h, *)$ be a subgroup of G. Then, it holds that $|H|$ divides $|G|$.*

Groups are one of several algebraic structure, among which we also consider rings and fields. The latter will be at the center of our discussion on cryptography.

Definition (Ring): Let R be a set, and let $+, *$ be two distinct binary operations on R's elements. We say that $(R, +, *)$ is a ring if

1. $(R, +)$ is an abelian group;
2. The operation $*$ is associative;
3. The operation $*$ distibutes over $+$; that is,

$$\forall a, b, c \in G, a * (b + c) = a * b + a * c.$$

Additionally, if R has a multiplicative unit, i.e., $\exists e' \in R$ such that $e' * a = a * e' = a, \forall a \in R$ and $*$ is commutative, we say that $(R, +, *)$ is a commutative ring with unity.

In what follows we denote by $\mathcal{K}^*$ the ring $\mathcal{K} \setminus \{0\}$.

Definition (Field): Given a commutative ring with unity $(R, +, *)$, we say that $(R, +, *)$ is a field if $(R^*, +, *)$ is commutative.

Similarly for groups, each field has an order given by the number of its elements. In these notes we mostly consider fields of order a prime and denote these by $\texttt{gf}\,(p)$, for some prime number p.

– 2 – Public-key cryptography.

Todays cryptography is almost entirely based on public-key schemas. Broadly speaking, each of the entities involved in the communication owns a public key e, available by everyone, and a private key d that must be kept secret. To each key a transformation is associated, respectively to encrypt and decrypt the message.

Here we focus on the most used public-key schemas, while in **??** we describe how such schemas are used to garantee the autenticity of messages. Each of the schemas we are about to discuss is based on the hardness of some mathematical problem. Also, we denote by A and B two entities that want to communicate, and by E some unauthorized user.

– 2.1 – RSA schema.

Named after its inventors, Rivest, Shamir, and Adleman, RSA is a public-key schema based on the hardness of integers factorization. The core idea is simple: choose two odd prime p and q and compute their product n. If m is the message to be transmitted, encode it using n and send the encoded message. Procedure RSA-Keygen describes how each entity involved computes its keys, while Procedure RSA-Encode-Decode shows how a message is encoded/decoded.

Procedure RSA-Keygen

Choose two odd prime numbers p and q and compute $n = pq$

Compute $l = (p-1)(q-1)$ and choose $1 \leq e < l$ such that $\gcd(e, l) = 1$

Using the extended Euclidean algorithm, find d such that $ed \equiv 1 \mod l$

Publish (n, e) as public key.

Here e is called *encryption key* and d is called *decryption key*, while n is called *modulo*.

Procedure RSA-Encode-Decode

Get a copy of A's public key ;

Convert your message as an integer $m \in [0, n-1]$

Cypher m as the integer $c = m^e \mod n$ and transmit c to A

`/* On the reciveing side: */`

Retrieve m by computing $c^{ed} \mod n$

– 2.2 – ElGamal schema.

The ElGamal schema is base on the hardness of the discrete logarithm, i.e., the problem of finding the logarithm for $a^n \mod n$, for some a, b and n.

Algorithmically speaking we preoceed as follow: we choose a cyclic group G generated by some integer γ such that $|G| = p$, with p a prime. The group $G^* = G \setminus \{0\}$ is considered. Once the group is choosed, we create the public-keys as in Procedure ElGamal-Keygen, and encode/decode the message using Procedure ElGamal-Encode-Decode

Procedure ElGamal-Keygen

Select a multiplicative group $\mathbb{Z}_p*$ generated by an element γ
Randomly choose an integer $a, 1 \leq a \leq p-2$ and compute $\gamma^a \mod p$
Use the triplet (p, γ, γ^a) as public-key, and a as the private one.

Procedure ElGamal-Encode-Decode

Get a copy of A's
Convert the message m as an integer in the range $[1, p-2]$
Randomly choose $k, 1 \leq k \leq p-2$, then compute $\delta = \gamma^k$ and $\epsilon = m(\gamma^a)^k$
Forward (δ, ϵ) to A
`/* On the receiving side: */`
Use the private key a to compute $\delta^{p-1-a} = \delta^{-a}$
Retrieve m by computing $\delta^{-a}\epsilon \mod p$

The correctness of the decoding is trivial. Since the group is cyclic, the product is commutative, meaning that $\delta^{-a}\epsilon = \gamma^{-ak}m\gamma ak = m$.

References.

[1] R. L. Rivest, A. Shamir, and L. Adleman. "A Method for Obtaining Digital Signatures and Public-key Cryptosystems". In: *Commun. ACM* 21.2 (Feb. 1978), pp. 120–126. DOI: 10.1145/359340.359342.